



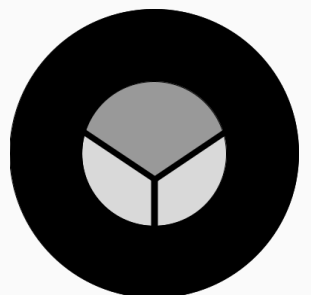
Down the River

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DE Intro to Statistics I

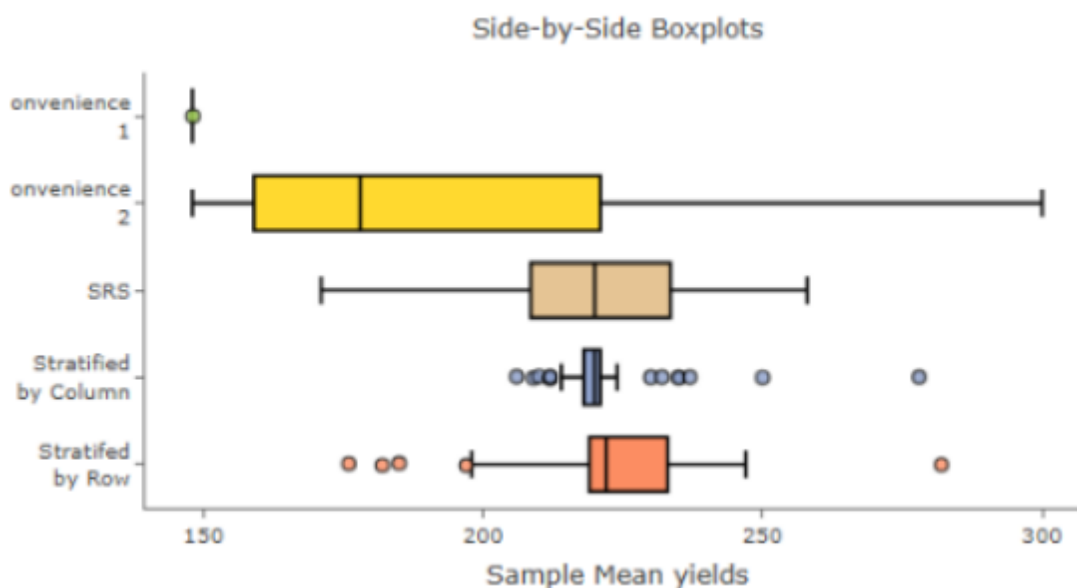
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“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton



Instructions: Use the graphical displays and statistical summaries to answer the questions. Recall, the farmer is going to use the results of a sample ($n = 10$) to determine whether it is worthwhile to harvest all plots ($N = 100$).

Mean yields	n	Mean	Std. Dev.	Min	Q1	Median	Q3	Max	IQR
convenience 1	137	148.00073	0.00854	148.00000	148.00000	148.00000	148.00000	148.10000	0.00000
convenience 2	137	195.95620	43.15514	148.00000	159.00000	178.00000	221.00000	300.00000	62.00000
SRS	137	220.68613	17.28981	171.00000	208.50000	220.00000	233.50000	258.00000	25.00000
Stratified by Column	137	220.59854	6.79150	206.00000	218.00000	220.00000	221.00000	278.00000	3.00000
Stratified by Row	136	223.82353	13.31437	176.00000	219.00000	222.00000	233.00000	282.00000	14.00000



0.0.1 Use the summary statistics to make a compelling case for why one design is better than the other two when comparing the SRS, Stratified by Column, and Stratified by Row sample designs.

Stratified by Column is the best design to use for the problem Rolling Down the River. Stratified by Column has a mean of 220.60, and a standard deviation of 6.79, while Stratified by Row's mean is 223.82, and its standard deviation is 13.3. Lastly, the Simple Random Sample yields a mean of 220.68, and a standard deviation of 17.29. While all three designs have similar means, the difference is all in the spread. While Stratified by Column has a lot of outliers, this is only due to the outlier fence rule; almost all of Column's outliers lie within the 75% of stratified by Row. Furthermore, since the IQR is low, 50% of the data in Stratified by Column lies in a very small spread (218-221), while SRS and Row lie in (208.5-233.5) and (219-233) respectively. By stratifying by column, each level gets equal representation.